

Ingestion Platform - Technical Reference

Overview

The ingestion platform converts enterprise source material - slide decks, PDFs, and workspace documents - into publication-ready reference documentation. Source material is normalised into chunked text with slide-accurate figure extraction so that every generated statement remains traceable to its origin.

Retrieval

Chunks are ranked by a hybrid scorer that fuses BM25 lexical matching with embedding similarity. Lexical scoring preserves precision on exact identifiers such as model names and ticket references; embedding similarity recovers paraphrased intent. The fused ranking assembles the grounded context window that constrains generation.

Generation

Requests are routed to a fast, default, or strong model according to the requested detail level. When the inference tier is unavailable, a deterministic structure-preserving composer produces a grounded document from the retrieved context and the run is recorded as degraded rather than failed.

Rendering

Output is rendered to PDF and DOCX with section headings, figure captions, and tables. Presentation artefacts from the source - slide numbers, action items, and meeting chatter - are removed during post-processing.

Measured Throughput

Configuration	Throughput (docs/min)	Median latency (s)
Baseline	18	4.2
Tuned	42	1.8

Configuration	Throughput (docs/min)	Median latency (s)
Cached	61	1.1

Telemetry

Every run emits a structured record of more than fifty KPIs covering retrieval quality, generation quality, content integrity, and operational cost. These records back both the live dashboard and the offline quality analysis.